

Umesh Karunarathna

IT Operations Engineer

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PROFESSIONAL SUMMARY

IT Operations professional with over 3 years of experience supporting enterprise infrastructure, end-user technologies, and business-critical IT services within a 24/7 environment. Experienced in Windows, Linux, Microsoft 365, Microsoft Entra ID, Active Directory, VMware, networking, and ITIL-based service management. Skilled in endpoint management, identity and access administration, technical troubleshooting, scripting with PowerShell, Python, and Bash, and process automation. Passionate about building secure, scalable IT operations through automation, documentation, and continuous improvement.

TECHNICAL SKILLS

Operating Systems: Windows Server 2019–2022, Windows 10/11, Linux, macOS

Endpoint & Collaboration: Windows Deployment, Hardware & Software Troubleshooting, Microsoft 365, Teams, SharePoint, Microsoft Office Suite, Remote Support, User Onboarding & Offboarding, Device Provisioning

Directory & Identity: Active Directory, Group Policy (GPO), Microsoft Entra ID (Azure AD), Azure AD Connect, Microsoft 365 Administration, Intune, User Provisioning & Access Management, Multi-Factor Authentication (MFA)

Networking: TCP/IP, DNS, DHCP, VPN, LAN/WAN, Routing & Switching, Cisco Meraki, Cisco ISE, Cisco ACI

Cloud & Virtualization: Microsoft Azure, AWS (EC2), VMware ESXi, VMware vSphere, Citrix Virtual Apps & Desktops

IT Service Management: ServiceNow, OP5, ITIL v4 Incident, Change & Request Management, System Monitoring

Security & Infrastructure: Palo Alto NGFW, Fortinet, Cisco ASA, Citrix ADC (NetScaler SDX), Vulnerability Assessment, Patch Compliance, Web Filtering Administration, Wazuh SIEM, Identity & Access Management

Scripting & Automation: PowerShell, Python, Bash/Shell Scripting, Process Automation, GitHub Copilot, ChatGPT

Professional Skills: Technical Troubleshooting, Root Cause Analysis, Technical Documentation, Process Improvement, Customer Service, Cross-Functional Collaboration, Problem Solving

PROFESSIONAL EXPERIENCE

Systems Engineer - Network Operations Center (NOC) IFS | Colombo, Sri Lanka | November 2020 - December 2023

Played a key role in building the NOC from the ground up; standardizing incident handling workflows, monitoring procedures, and reporting structures that improved operational efficiency and reduced response times across a 24/7 global network environment.

- Maintained continuous availability of global IFS network infrastructure across 3+ regions on a 24/7 shift roster, triaging and resolving 15+ incidents daily within defined SLA targets.
- Configured and managed enterprise firewalls (Fortinet, Palo Alto NGFW, Cisco ASA), DNS/DHCP servers, and Citrix ADC/NetScaler SDX appliances supporting 1,000+ users.
- Administered Palo Alto Networks firewalls through Panorama, along with Cisco ASA and Fortinet firewalls, performing firewall policy changes, NAT, VPN troubleshooting, security policy management, and firewall administration.
- Designed and deployed network infrastructure for 5+ new branch offices, including VPN setup, routing configuration, and end-to-end connectivity validation.
- Administered hybrid identity and device management using Active Directory (user provisioning, GPOs), Microsoft EntraID (Azure AD), and Intune for access control, policy enforcement, and endpoint compliance.
- Performed scheduled upgrades on network devices and firewalls with zero unplanned downtime, following ITIL v4 change management procedures.
- Monitored real-time network performance using Nagios-based tools across 200+ devices, proactively identifying and resolving incidents before business impact.
- Conducted regular vulnerability scans across network infrastructure and drove remediation of identified risks within SLA windows.

- Managed datacenter infrastructure using Cisco ACI and coordinated network relocation projects with minimal service disruption.
- Provided technical support for end users by troubleshooting hardware, software, network connectivity, account access, and Microsoft 365 issues while collaborating with infrastructure and security teams to restore services within SLA targets.
- Engaged with 10+ external vendors and partners to resolve third-party hardware and software issues, maintaining clear communication and escalation records.

Student Assistant - Computer Labs Douglas College | New Westminster, BC | September 2024 – May 2026 (*Part-Time*)

- Provided first-line technical support for Windows and Linux workstations, software applications, virtualization platforms, and networking labs, assisting students and faculty with troubleshooting and issue resolution.
- Supported 15+ students per shift with hands-on troubleshooting of OS, virtualization, and networking concepts.

Sports Advisor - Fitness & Team Sports Decathlon | Burnaby, BC | May 2025 - Present (*Part-Time*)

- Deliver customer-focused service to 30+ customers daily, assessing needs and recommending solutions.
- Collaborate in a fast-paced team environment, demonstrating adaptability, communication, and time management under pressure.

EDUCATION

Post-Baccalaureate Diploma - Computer and Information Systems

Major: Cybersecurity | April 2026 | Douglas College | Cumulative GPA: 4.11 / 4.33 | New Westminster, BC

Relevant Coursework: Operating Systems, Virtualization & Computer Networks, System Analysis, Evidence Imaging, Scripting for Cybersecurity, Cloud Infrastructure, Advanced Integrated Software Development (Java), Network Cybersecurity, Cloud Cybersecurity, Mobile Cybersecurity.

Key Projects:

- **Enterprise Active Directory & Windows Server Administration:** Designed and administered a Windows Server 2022 enterprise environment featuring Active Directory Domain Services (AD DS), DNS, DHCP, Organizational Units (OUs), Group Policy (GPO), shared folders with NTFS permissions, mapped network drives, Windows 11 domain-joined clients, and PowerShell automation for user provisioning and administration.
- **AI-Powered SOC Assistant:** Built a locally hosted LLM-based SOC assistant using Ollama (Mistral), AWS EC2 instances, and Wazuh SIEM; integrated Ubuntu and Kali Linux endpoints to simulate real-world threat detection and log analysis workflows.
- **Enterprise Security Hardening & Monitoring:** Hardened and monitored an Ubuntu server environment covering account and password policies, network security, kernel hardening, logging, and auditing using industry-standard frameworks.

Bachelor of Science (Honours) - Computer Networks

First Class Honours | Graduated July 2020 | University of Plymouth | Plymouth, UK

Key Projects:

- **Dynamic Bandwidth Management:** Designed an adaptive network bandwidth allocation algorithm based on real-time user behaviour analysis.
- **Centralized Network Management:** Configured Group Policies and user management using Microsoft Active Directory in a simulated enterprise environment.

CERTIFICATIONS & LICENSES

Certified in Cybersecurity (CC) ISC2 | May 2024

ITIL v4 Foundation AXELOS | August 2021

Introduction to Cybersecurity Cisco Networking Academy | May 2020